E10 Multiple Reactions in a PFR

The following reactions are carried out in the gas phase in a 400-dm^3 PFR at 325 K:

$$\frac{r_{1A}}{-3} = \frac{r_{1B}}{1} \quad \therefore \quad r_{1B} = -\frac{1}{3} r_{1A}$$

3A --> B + C	$-r_{1A} = k_{1A}C_A$	k_{1A} = 7.0 min^{-1}
2C + A --> 3D	$r_{2D} = k_{2D}C_C^2C_A$	k_{2D} = 3.0 dm^6/mol^2*min
4D + 3C --> 3E	$r_{3E} = k_{3E}C_CC_D$	k_{3E} = 2.0 dm^3/mol*min

A is fed at a concentration of 0.2 mol/dm^3, and the inlet volumetric flowrate is 100 dm^3/min. Plot the concentrations of A, B, C, D, and E in the reactor and determine $\tilde{S}_{D/E}$ and $\tilde{S}_{C/D}$.

<u>mole balances</u>

$$\frac{dF_A}{dV} = r_A \qquad \frac{dF_B}{dV} = r_B \qquad \frac{dF_C}{dV} = r_C$$

$$\frac{dF_D}{dV} = r_D \qquad \frac{dF_E}{dV} = r_E$$

<u>net rates of generation</u>

$$r_A = r_{1A} + r_{2A}$$

$$r_B = r_{1B}$$

$$r_C = r_{1C} + r_{2C} + r_{3C}$$

$$r_D = r_{2D} + r_{3D}$$

$$r_E = r_{3E}$$

relative rates

$$r_{1B} = -\frac{1}{3} r_{1A}$$

$$r_{1C} = r_{1B}$$

$$r_{2C} = -\frac{2}{3} r_{2D}$$

$$r_{2A} = -\frac{1}{3} r_{2D}$$

$$r_{3D} = -\frac{4}{3} r_{3E}$$

$$r_{3C} = -r_{3E}$$

concentration equations :

$$C_j = C_{T0}\left(\frac{F_j}{F_T}\right)\underbrace{\cancel{\left(\frac{P}{P_0}\right)}}_{=1,\text{ no pressure drop}}\overbrace{\cancel{\left(\frac{T}{T_0}\right)}}^{=1,\text{ isothermal}}$$

$$F_T = F_A + F_B + F_C + F_D + F_E$$

$$C_A = C_{T0}\frac{F_A}{F_T} \qquad C_B = C_{T0}\frac{F_B}{F_T} \qquad C_C = \frac{F_C}{F_T} C_{T0}$$

$$C_D = C_{T0}\frac{F_D}{F_T} \qquad C_E = C_{T0}\frac{F_E}{F_T}$$